

A Pannon Egyetem gyakorlatorientált infrastrukturális- és készségfejlesztési reformja

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NUMERICAL ANALYSIS

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Numerical Analysis

6 Interpolation

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6.1 Lagrange Interpolation

Given pairwise different points

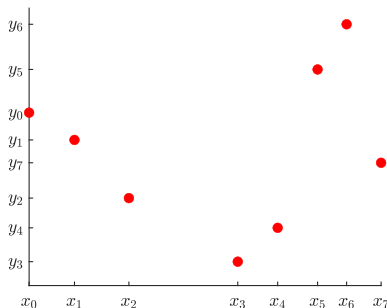
$$x_0, x_1, \dots, x_n \in [a, b],$$

the so-called **mesh points** or **node points**, and corresponding function values

$$y_0, y_1, \dots, y_n.$$

The basic problem of interpolation is to find a function g from a certain class of functions which **interpolates** the given data, i.e., satisfies relations

$$g(x_i) = y_i, \quad i = 0, 1, \dots, n.$$



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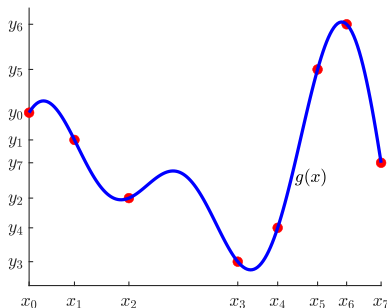
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We are looking for a polynomial L_n of degree at most n which satisfies

$$L_n(x_i) = y_i, \quad i = 0, 1, \dots, n. \quad (1)$$

This problem is called **Lagrange interpolation**. We show that this problem has a unique solution. The solution L_n of this problem is called **Lagrange interpolating polynomial**, or shortly, **Lagrange polynomial**.

For $k = 0, 1, \dots, n$ we define the polynomial of degree n by

$$l_k(x) := \frac{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)}.$$

The polynomials $l_0, \dots, l_n$ are called **Lagrange basis polynomials of degree n** . It follows from the definition that

$$l_k(x_i) = \begin{cases} 1, & \text{if } k = i, \\ 0, & \text{if } k \neq i. \end{cases}$$

It follows that the polynomial

$$L_n(x) := \sum_{k=0}^n y_k l_k(x)$$

is of degree at most n , and it solves the Lagrange interpolation problem (1).

Theorem:

The Lagrange interpolating problem has a unique solution which can be given by

$$L_n(x) = \sum_{k=0}^n y_k \frac{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)}$$

Proof

Suppose L_n and $\tilde{L}_n$ are polynomials of degree at most n , and both are solutions of problem (1). We define the function

$$P(x) := L_n(x) - \tilde{L}_n(x).$$

Then P is a polynomial of degree at most n , and

$$P(x_i) = 0 \quad \text{for all } i = 0, 1, \dots, n,$$

i.e., P has $n + 1$ different roots. But then the Fundamental theorem of algebra yields that P is identically equal to 0, i.e.,

$$L_n = \tilde{L}_n.$$

Example

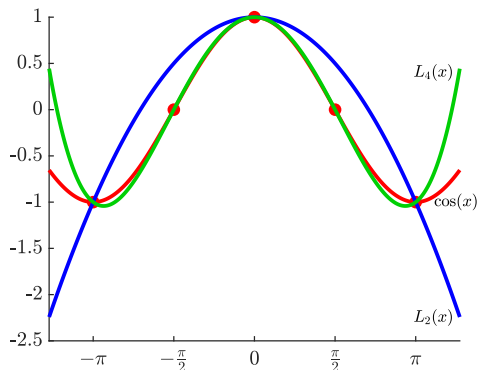
Consider the given data

x_i	-1	1	2	3
y_i	-2	0	-2	2

Find the Lagrange polynomial which interpolates the above data.

We get

$$\begin{aligned}
 L_3(x) &= -2 \frac{(x-1)(x-2)(x-3)}{(-1-1)(-1-2)(-1-3)} + 0 \frac{(x+1)(x-2)(x-3)}{(1+1)(1-2)(1-3)} \\
 &\quad - 2 \frac{(x+1)(x-1)(x-3)}{(2+1)(2-1)(2-3)} + 2 \frac{(x+1)(x-1)(x-2)}{(3+1)(3-1)(3-2)} \\
 &= x^3 - 3x^2 + 2.
 \end{aligned}$$



Lagrange interpolation of the function $\cos x$ using the mesh points $-\pi, 0, \pi$ and the mesh points $-\pi, -\pi/2, 0, \pi/2, \pi$, respectively

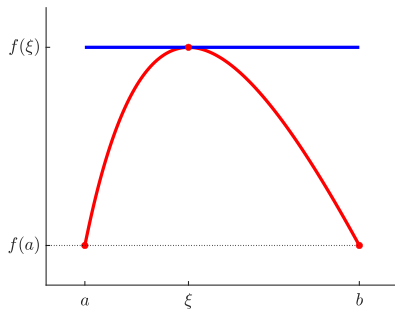
Theorem (Rolle's Theorem):

Let $f \in C^1[a, b]$ and

$$f(a) = f(b).$$

Then there exists $\xi \in (a, b)$ such that

$$f'(\xi) = 0.$$



Theorem (Generalized Rolle's Theorem):

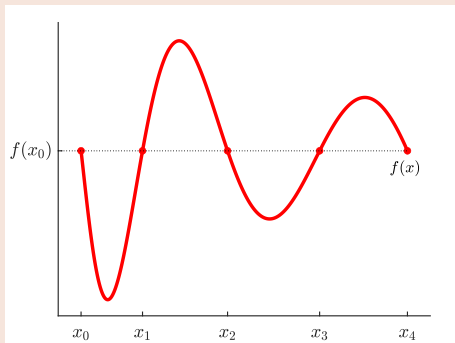
Let $f \in C^n[a, b]$, $a \leq x_0 < x_1 < \dots < x_n \leq b$, and suppose

$$f(x_0) = f(x_1) = \dots = f(x_n).$$

Then there exists $\xi \in (x_0, x_n)$ such that

$$f^{(n)}(\xi) = 0.$$

Proof



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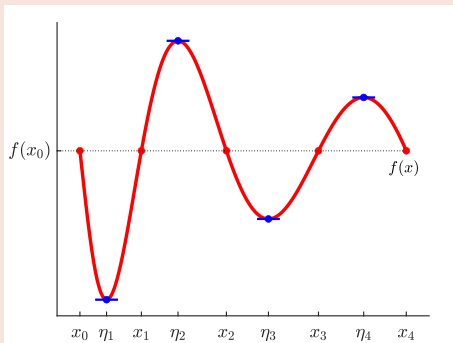
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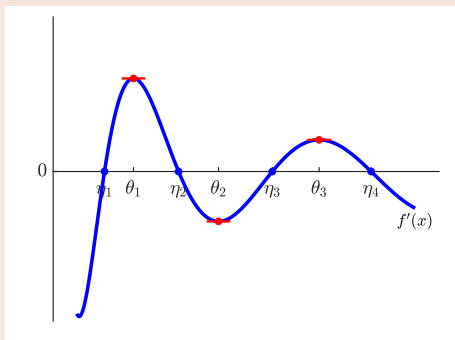
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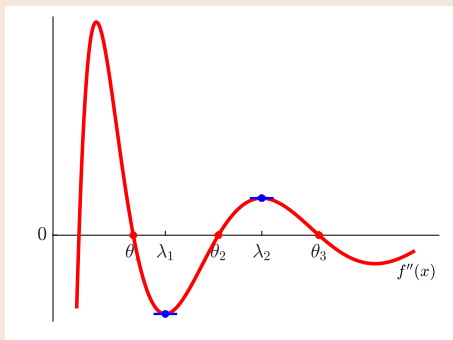
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Proof



Theorem (Generalized Rolle's Theorem):

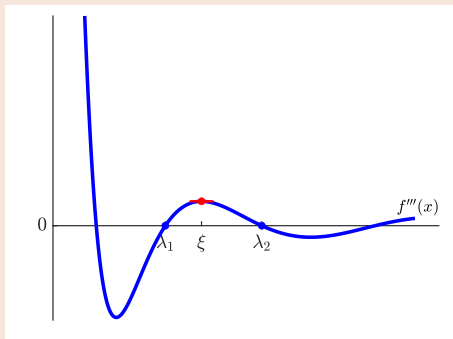
Let $f \in C^n[a, b]$, $a \leq x_0 < x_1 < \dots < x_n \leq b$, and suppose

$$f(x_0) = f(x_1) = \dots = f(x_n).$$

Then there exists $\xi \in (x_0, x_n)$ such that

$$f^{(n)}(\xi) = 0.$$

Proof



Theorem:

Let $f \in C^{n+1}[a, b]$, $x_i \in [a, b]$ ($i = 0, \dots, n$) be pairwise distinct mesh points and $y_i = f(x_i)$ ($i = 0, \dots, n$). Let $L_n(x)$ be the corresponding Lagrange interpolating polynomial. Then for every $x \in [a, b]$ there exists $\xi = \xi(x) \in \langle x, x_0, x_1, \dots, x_n \rangle$ such that

$$f(x) = L_n(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)(x - x_1) \cdots (x - x_n).$$

Proof

Fix $x \in (a, b)$ such that $x \neq x_i$ for all $i = 0, \dots, n$, and consider

$$g(t) := f(t) - L_n(t) - \frac{(t - x_0) \cdots (t - x_n)}{(x - x_0) \cdots (x - x_n)} (f(x) - L_n(x)).$$

Clearly, $g \in C^{n+1}$, and

$$g(x) = g(x_0) = g(x_1) = \cdots = g(x_n) = 0.$$

Then the generalized Rolle's Theorem yields that there exists a number $\xi \in \langle x, x_0, \dots, x_n \rangle$ such that

$$g^{(n+1)}(\xi) = 0.$$

Since L_n is a polynomial of degree at most n , its $(n+1)$ -st-order derivative is identically 0, so

$$g^{(n+1)}(t) = f^{(n+1)}(t) - \frac{(n+1)!}{(x - x_0) \cdots (x - x_n)} (f(x) - L_n(x)).$$

This gives the statement with $t = \xi$.

Now we consider the case when the mesh points are equidistant, i.e., $x_i = x_0 + ih$. The truncation error of the interpolation can be estimated by

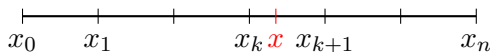
$$|f(x) - L_n(x)| \leq \frac{M_{n+1}}{(n+1)!} |(x - x_0) \cdots (x - x_n)|$$

where

$$M_{n+1} = \sup\{|f^{(n+1)}(t)| : t \in [x_0, x_n]\}.$$

Suppose $x \in (x_k, x_{k+1})$ for some $0 \leq k < n$. Then we have

$$|(x - x_k)(x - x_{k+1})| \leq \frac{h^2}{4},$$



and so

$$\begin{aligned}
 \prod_{i=0}^n |x - x_i| &\leq \frac{h^2}{4} \prod_{i=0}^{k-1} (x - x_i) \prod_{i=k+2}^n (x_i - x) \\
 &\leq \frac{h^2}{4} \prod_{i=0}^{k-1} (x_{k+1} - x_i) \prod_{i=k+2}^n (x_i - x_k) \\
 &= \frac{h^{n+1}}{4} \prod_{i=0}^{k-1} (k+1-i) \prod_{i=k+2}^n (i-k) \\
 &= \frac{h^{n+1}}{4} (k+1)!(n-k)! \\
 &\leq \frac{h^{n+1}}{4} n!
 \end{aligned}$$

Theorem:

Let $f \in C^{n+1}[a, b]$,

$$x_i = a + i \frac{b-a}{n} \quad (i = 0, \dots, n)$$

and $y_i = f(x_i)$ ($i = 0, \dots, n$). Let $x \in [a, b]$. Then

$$|f(x) - L_n(x)| \leq \frac{M_{n+1}}{4(n+1)} \left(\frac{b-a}{n} \right)^{n+1},$$

where $M_{n+1} := \sup\{|f^{(n+1)}(x)| : x \in [a, b]\}$.

Example

Consider $f(x) = \cos x$. According to the previous theorem it follows for $x \in [-\pi, \pi]$

$$|f(x) - L_2(x)| \leq \frac{1}{12}\pi^3 \approx 2.5839, \text{ and } |f(x) - L_4(x)| \leq \frac{1}{20} \left(\frac{\pi}{2}\right)^5 \approx 0.4782.$$

Let $f: [a, b] \times [c, d] \rightarrow \mathbb{R}$, and consider the division of $[a, b]$ and $[c, d]$ by $a = x_0 < x_1 < \dots < x_n = b$ and $c = y_0 < y_1 < \dots < y_m = d$. Let

$$z_{ij} = f(x_i, y_j), \quad i = 0, \dots, n, \quad j = 0, \dots, m.$$

We define the following two-variable polynomial to interpolate the given data:

$$L_{n,m}(x, y) := \sum_{i=0}^n \sum_{j=0}^m z_{ij} l_i(x) \tilde{l}_j(y),$$

where l_i and $\tilde{l}_j$ are the Lagrange basis polynomials of degree n and m , respectively, corresponding to the mesh points

$a = x_0 < x_1 < \dots < x_n = b$ and $c = y_0 < y_1 < \dots < y_m = d$. The function $L_{n,m}$ satisfies

$$L_{n,m}(x_i, y_j) = z_{ij}, \quad i = 0, \dots, n, \quad j = 0, \dots, m.$$

If x is fixed, then $L_{n,m}(x, \cdot)$ is a polynomial of degree at most m .

Conversely, if y is fixed, then $L_{n,m}(\cdot, y)$ is a polynomial of degree at most n .

Example

Consider the following given function values:

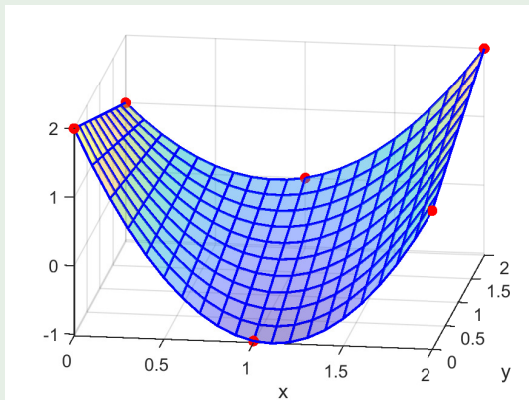
(x_i, y_j)	(0, 0)	(1, 0)	(2, 0)	(0, 2)	(1, 2)	(2, 2)
z_{ij}	2	-1	1	1	0	2

Then

$$\begin{aligned}
 L_{2,1}(x, y) &= 2 \frac{(x-1)(x-2)}{(0-1)(0-2)} \frac{y-2}{0-2} - \frac{x(x-2)}{1(1-2)} \frac{y-2}{0-2} + \frac{x(x-1)}{2(2-1)} \frac{y-2}{0-2} \\
 &\quad + \frac{(x-1)(x-2)}{(0-1)(0-2)} \frac{y}{2} + 0 \frac{x(x-2)}{1(1-2)} \frac{y}{2} + 2 \frac{x(x-1)}{2(2-1)} \frac{y}{2} \\
 &= -\frac{1}{2}x^2y + \frac{5}{2}x^2 + \frac{3}{2}xy - \frac{11}{2}x - \frac{1}{2}y + 2.
 \end{aligned}$$

This is of second-order in x , and first-order in y . The graph of the polynomial can be seen in the next figure.

Example cont.



Two-variable Lagrange interpolation

6.2 Divided Differences

Given a function $f: [a, b] \rightarrow \mathbb{R}$ and pairwise different mesh points $x_i \in [a, b]$ ($i = 0, \dots, n$). Then the **zeroth divided difference** of the function f at the point x_0 is defined by

$$f[x_0] := f(x_0)$$

The **first divided difference** of the function f at the points x_0, x_1 is the number

$$f[x_0, x_1] := \frac{f[x_1] - f[x_0]}{x_1 - x_0},$$

i.e.,

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}.$$

In general, the **n th divided difference** of the function f relative to the points $x_0, x_1, \dots, x_n$ is defined by

$$f[x_0, x_1, \dots, x_n] := \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}.$$

Theorem:

Let x_i ($i = 0, 1, \dots, n$) be pairwise different mesh points. Then

$$f[x_0, x_1, \dots, x_n] = \sum_{i=0}^n \frac{f(x_i)}{(x_i - x_0) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_n)}.$$

Proof

We prove the statement using mathematical induction with respect to n . For $n = 0$ the statement is obvious. (In this case in the denominator we have the “empty product”, which, by definition, equals to 1.) Suppose the statement holds for n , and consider the $n + 1$ -th divided difference $f[x_0, x_1, \dots, x_{n+1}]$.

Proof cont.

$$\begin{aligned}
& f[x_0, x_1, \dots, x_{n+1}] \\
&= \frac{f[x_1, x_2, \dots, x_{n+1}] - f[x_0, x_1, \dots, x_n]}{x_{n+1} - x_0} \\
&= \frac{1}{x_{n+1} - x_0} \left\{ \sum_{i=1}^{n+1} \frac{f(x_i)}{(x_i - x_1) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_{n+1})} \right. \\
&\quad \left. - \sum_{i=0}^n \frac{f(x_i)}{(x_i - x_0) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_n)} \right\} \\
&= \frac{1}{x_{n+1} - x_0} \left\{ \frac{f(x_{n+1})}{(x_{n+1} - x_1) \cdots (x_{n+1} - x_n)} - \frac{f(x_0)}{(x_0 - x_1) \cdots (x_0 - x_n)} \right. \\
&\quad \left. + \sum_{i=1}^n \frac{f(x_i)}{(x_i - x_1) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_n)} \right. \\
&\quad \left. \cdot \left(\frac{1}{x_i - x_{n+1}} - \frac{1}{x_i - x_0} \right) \right\} \\
&= \sum_{i=0}^{n+1} \frac{f(x_i)}{(x_i - x_0) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_{n+1})}.
\end{aligned}$$

The previous result has some immediate consequences.

Corollary

The divided differences are independent of the order of the mesh points.

Corollary

If the function f is continuous, then the divided differences depend continuously on the mesh points.

Suppose f is differentiable. The function

$$x_1 \mapsto f[x_0, x_1]$$

is continuous for $x_1 \neq x_0$. We get

$$\lim_{x_1 \rightarrow x_0} f[x_0, x_1] = \lim_{x_1 \rightarrow x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0} = f'(x_0).$$

Therefore we define the first divided difference relative to equal mesh points by

$$f[x_0, x_0] := f'(x_0).$$

With this definition the function

$$x_1 \mapsto f[x_0, x_1]$$

is extended continuously for $x_1 = x_0$.

6.3 Newton's Divided Difference Formula

Suppose function values $y_i = f(x_i)$ are given for $i = 0, 1, \dots, n$. First consider the relation

$$L_n(x) = L_0(x) + (L_1(x) - L_0(x)) + (L_2(x) - L_1(x)) + \dots + (L_n(x) - L_{n-1}(x)).$$

By definition, $L_0(x) = f(x_0)$. Consider the difference $L_i(x) - L_{i-1}(x)$. It is a polynomial of degree at most i , and since L_i and L_{i-1} both satisfy the interpolating equations at $x_0, \dots, x_{i-1}$, we have

$$L_i(x_j) - L_{i-1}(x_j) = f(x_j) - f(x_j) = 0, \quad j = 0, 1, \dots, i-1.$$

But then the Fundamental theorem of algebra yields

$$L_i(x) - L_{i-1}(x) = a_i(x - x_0)(x - x_1) \cdots (x - x_{i-1}),$$

where $a_i \in \mathbb{R}$. If we substitute $x = x_i$ into this relation, we get

$$\begin{aligned} f(x_i) - \sum_{k=0}^{i-1} f(x_k) \frac{(x_i - x_0) \cdots (x_i - x_{k-1})(x_i - x_{k+1}) \cdots (x_i - x_{i-1})}{(x_k - x_0) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_{i-1})} \\ = a_i(x_i - x_0) \cdots (x_i - x_{i-1}). \end{aligned}$$

So from this we get for a_i that

$$\begin{aligned}
 a_i &= \frac{f(x_i)}{(x_i - x_0) \cdots (x_i - x_{i-1})} - \frac{1}{(x_i - x_0) \cdots (x_i - x_{i-1})} \\
 &\quad \times \sum_{k=0}^{i-1} f(x_k) \frac{(x_i - x_0) \cdots (x_i - x_{k-1})(x_i - x_{k+1}) \cdots (x_i - x_{i-1})}{(x_k - x_0) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_{i-1})} \\
 &= \sum_{k=0}^i \frac{f(x_k)}{(x_k - x_0) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_i)} \\
 &= f[x_0, x_1, \dots, x_i].
 \end{aligned}$$

Therefore the Lagrange interpolating polynomial can be written as

$$L_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \cdots \\ + f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \cdots (x - x_{n-1}). \quad (2)$$

The polynomial given by (2) is called **Newton's Divided Difference formula** or shortly **Newton polynomial**.

It is easy to add a new mesh point to the formula, we have the simple correction term:

$$L_{n+1}(x) = L_n(x) + f[x_0, x_1, \dots, x_{n+1}](x - x_0) \cdots (x - x_n).$$

Algorithm: Computation of the coefficients of the Newton polynomial

INPUT: n - number of mesh points – 1
 x_i , ($i = 0, 1, \dots, n$) - mesh points
 y_i , ($i = 0, 1, \dots, n$) - function values

OUTPUT: a_i , ($i = 0, 1, \dots, n$) - coefficients of the Newton polynomial,
 where a_i is the coefficient of the i th-order term

```

for  $i = 0, 1, \dots, n$  do
     $a_i \leftarrow y_i$ 
end do
for  $j = 1, 2, \dots, n$  do
    for  $i = n, n - 1, \dots, j$  do
         $a_i \leftarrow (a_i - a_{i-1}) / (x_i - x_{i-j})$ 
    end do
end do
output( $a_0, a_1, \dots, a_n$ )
  
```

Algorithm: Evaluation of the Newton polynomial

INPUT: n - number of mesh points – 1
 $x_i, (i = 0, 1, \dots, n)$ - mesh points
 $a_i, (i = 0, 1, \dots, n)$ - coefficients of the Newton polynomial
 x - the value where we evaluate the Newton polynomial

OUTPUT: y - function value of the Newton polynomial at x

```

 $y \leftarrow a_n$ 
for  $i = n - 1, n - 2, \dots, 0$  do
     $y \leftarrow y(x - x_i) + a_i$ 
end do
output( $y$ )
  
```

Computation of the divided differences by hand

x_0	$f(x_0)$				
x_1	$f(x_1)$	$f[x_0, x_1]$			
x_2	$f(x_2)$	$f[x_1, x_2]$	$f[x_0, x_1, x_2]$		
x_3	$f(x_3)$	$f[x_2, x_3]$	$f[x_1, x_2, x_3]$	$\ddots$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$		
x_n	$f(x_n)$	$f[x_{n-1}, x_n]$	$f[x_{n-2}, x_{n-1}, x_n]$	$\cdots$	$f[x_0, x_1, \dots, x_n]$

Example

Consider again the previous example:

x_i	-1	1	2	3
y_i	-2	0	-2	2

We compute $L_3(x)$ in Newton's Divided Difference form, and we evaluate $L_3(0)$. First we compute the table of divided differences:

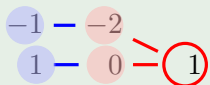
-1	-2			
1	0	×		
2	-2	×	×	
3	2	×	×	×

Example

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We compute $L_3(x)$ in Newton's Divided Difference form, and we evaluate $L_3(0)$. First we compute the table of divided differences:



$$\begin{array}{ccccc}
 -1 & - & -2 & & \\
 1 & - & 0 & = & 1 \\
 2 & -2 & \times & \times & \\
 3 & 2 & \times & \times & \times
 \end{array}$$

$$\frac{0 - (-2)}{1 - (-1)} = 1$$

Example

Consider again the previous example:

x_i	-1	1	2	3
y_i	-2	0	-2	2

We compute $L_3(x)$ in Newton's Divided Difference form, and we evaluate $L_3(0)$. First we compute the table of divided differences:

$$\begin{array}{ccccccc}
 & -1 & & -2 & & & \\
 & \textcircled{1} & \text{---} & \textcircled{0} & & 1 & \\
 & \textcircled{2} & \text{---} & \textcircled{-2} & \text{---} & \textcircled{-2} & \times \\
 & 3 & & 2 & & \times & \times & \times & \frac{-2-0}{2-1} = -2
 \end{array}$$

Example

Consider again the previous example:

x_i	-1	1	2	3
y_i	-2	0	-2	2

We compute $L_3(x)$ in Newton's Divided Difference form, and we evaluate $L_3(0)$. First we compute the table of divided differences:

$$\begin{array}{ccccccc}
 -1 & & -2 & & & & \\
 1 & & 0 & & 1 & & \\
 2 & \text{---} & -2 & & -2 & \times & \\
 3 & \text{---} & 2 & \text{---} & 4 & \times & \times & \frac{2-(-2)}{3-2} = 4
 \end{array}$$

Example

Consider again the previous example:

x_i	-1	1	2	3
y_i	-2	0	-2	2

We compute $L_3(x)$ in Newton's Divided Difference form, and we evaluate $L_3(0)$. First we compute the table of divided differences:

$$\begin{array}{cccc}
 -1 & -2 & & \\
 1 & 0 & 1 & \\
 2 & -2 & -2 & -1 \\
 3 & 2 & 4 & \times
 \end{array}
 \times \frac{-2 - (-2)}{2 - (-1)} = -1$$

Example

Consider again the previous example:

x_i	-1	1	2	3
y_i	-2	0	-2	2

We compute $L_3(x)$ in Newton's Divided Difference form, and we evaluate $L_3(0)$. First we compute the table of divided differences:

$$\begin{array}{ccccccc}
 & -1 & & -2 & & & \\
 1 & - & 0 & & 1 & & \\
 2 & & -2 & & -2 & & -1 \\
 3 & - & 2 & - & 4 & - & 3
 \end{array} \times \frac{4 - (-2)}{3 - 1} = 3$$

Example

Consider again the previous example:

x_i	-1	1	2	3
y_i	-2	0	-2	2

We compute $L_3(x)$ in Newton's Divided Difference form, and we evaluate $L_3(0)$. First we compute the table of divided differences:

$$\begin{array}{cccc} -1 & -2 & & \\ 1 & 0 & 1 & \\ 2 & -2 & -2 & -1 \\ 3 & 2 & 4 & 3 & 1 \end{array}$$

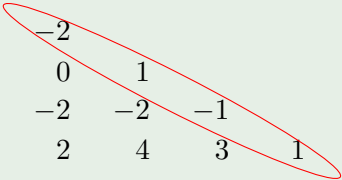
$$\frac{3 - (-1)}{3 - (-1)} = 1$$

Example

Consider again the previous example:

x_i	-1	1	2	3
y_i	-2	0	-2	2

We compute $L_3(x)$ in Newton's Divided Difference form, and we evaluate $L_3(0)$. First we compute the table of divided differences:



-1	-2			
1	0	1		
2	-2	-2	-1	
3	2	4	3	1

This yields that

$L_3(x) = -2 + (x + 1) - (x + 1)(x - 1) + (x + 1)(x - 1)(x - 2)$,
and so $L_3(0) = 2$. We can simplify this formula of L_3 and we get the
same form of the polynomial as earlier: $L_3(x) = x^3 - 3x^2 + 2$.

Theorem:

Let $x_i \in (a, b)$ ($i = 0, \dots, n$) be pairwise different mesh points and $y_i = f(x_i)$ ($i = 0, \dots, n$). Let $L_n(x)$ be the corresponding n th degree Lagrange interpolating polynomial. Then

$$f(x) = L_n(x) + f[x_0, x_1, \dots, x_n, x](x - x_0)(x - x_1) \cdots (x - x_n).$$

Proof

Fix $x \in (a, b)$ which is different from each mesh points. (If $x = x_i$ for some i , then the statement is clearly true.) Add x to the mesh points together with the function value $f(x)$. Let L_{n+1} be the Lagrange interpolation polynomial corresponding to the extended data set. Then we have

$$L_{n+1}(t) = L_n(t) + f[x_0, x_1, \dots, x_n, x](t - x_0) \cdots (t - x_n).$$

Now substitution $t = x$ proves the statement, since $f(x) = L_{n+1}(x)$.

Corollary

If $f \in C^n[a, b]$ and x_i ($i = 0, \dots, n$) are pairwise different mesh points, then there exists $\xi \in \langle x_0, x_1, \dots, x_n \rangle$ such that

$$f[x_0, x_1, \dots, x_n] = \frac{1}{n!} f^{(n)}(\xi).$$

6.4 Hermite Interpolation

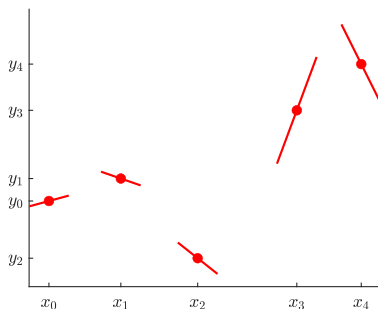
Let f be a differentiable function, and given mesh points x_i ($i = 0, \dots, n$). The **Hermite interpolation problem**: find a polynomial

$$g(x) = c_0 + c_1x + \dots + c_mx^m$$

which interpolates not only the function values $y_i = f(x_i)$, but also the derivative values $y'_i := f'(x_i)$.

Therefore we are looking for g which satisfies the interpolation conditions

$$g(x_i) = y_i, \quad g'(x_i) = y'_i, \quad i = 0, 1, \dots, n.$$



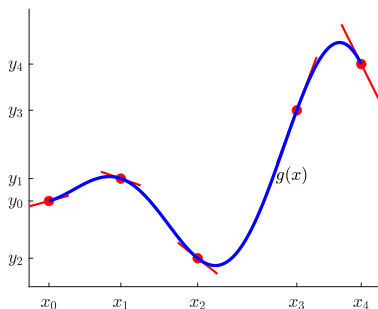
Let f be a differentiable function, and given mesh points x_i ($i = 0, \dots, n$). The **Hermite interpolation problem**: find a polynomial

$$g(x) = c_0 + c_1x + \dots + c_mx^m$$

which interpolates not only the function values $y_i = f(x_i)$, but also the derivative values $y'_i := f'(x_i)$.

Therefore we are looking for g which satisfies the interpolation conditions

$$g(x_i) = y_i, \quad g'(x_i) = y'_i, \quad i = 0, 1, \dots, n.$$



In the formula of the polynomial

$$g(x) = c_0 + c_1x + \cdots + c_mx^m$$

there are $m + 1$ parameters, and the interpolation conditions specify $2(n + 1)$ conditions. So we expect that the Hermite interpolation problem has a unique solution in the class of polynomials with degree at most

$$m = 2n + 1.$$

The solution of the Hermite interpolation problem is called **Hermite interpolation polynomial** or shortly **Hermite polynomial**, and it is denoted by

$$H_{2n+1}.$$

Later we use higher order divided differences where two consecutive mesh points can be equal: $f[x_0, x_0, x_1, x_1, \dots, x_n, x_n]$, where $x_0, \dots, x_n$ are pairwise different mesh points. Its definition is the usual recursion:

$$\begin{aligned} f[x_0, x_0, x_1, x_1, \dots, x_n, x_n] \\ = \frac{f[x_0, x_1, x_1, \dots, x_n, x_n] - f[x_0, x_0, x_1, x_1, \dots, x_n]}{x_n - x_0}. \end{aligned}$$

For example,

$$\begin{aligned} f[x_0, x_0, x_1] &= \frac{f[x_0, x_1] - f[x_0, x_0]}{x_1 - x_0} \\ f[x_0, x_1, x_1] &= \frac{f[x_1, x_1] - f[x_0, x_1]}{x_1 - x_0} \end{aligned}$$

Theorem:

The Hermite interpolation problem has a unique solution in the class of polynomials with degree at most $(2n + 1)$, which is given by

$$\begin{aligned}
 H_{2n+1}(x) &= f[x_0] + f[x_0, x_0](x - x_0) + f[x_0, x_0, x_1](x - x_0)^2 \\
 &\quad + f[x_0, x_0, x_1, x_1](x - x_0)^2(x - x_1) \\
 &\quad + f[x_0, x_0, x_1, x_1, x_2](x - x_0)^2(x - x_1)^2 \\
 &\quad + f[x_0, x_0, x_1, x_1, x_2, x_2](x - x_0)^2(x - x_1)^2(x - x_2) + \cdots \\
 &\quad + f[x_0, x_0, x_1, x_1, \dots, x_n, x_n](x - x_0)^2 \cdots (x - x_{n-1})^2(x - x_n).
 \end{aligned} \tag{3}$$

Moreover, the truncation error is

$$f(x) - H_{2n+1}(x) = f[x_0, x_0, \dots, x_n, x_n, x](x - x_0)^2 \cdots (x - x_n)^2. \tag{4}$$

Proof

First we discuss the uniqueness of the Hermite polynomial. Suppose H_{2n+1} and $\tilde{H}_{2n+1}$ are polynomials of degree at most $(2n+1)$ which both satisfy the equations of the Hermite interpolation problem. Then

$$P := H_{2n+1} - \tilde{H}_{2n+1}$$

is a polynomial of degree at most $(2n+1)$ which satisfy

$$P(x_i) = H_{2n+1}(x_i) - \tilde{H}_{2n+1}(x_i) = f(x_i) - f(x_i) = 0$$

and

$$P'(x_i) = H'_{2n+1}(x_i) - \tilde{H}'_{2n+1}(x_i) = f'(x_i) - f'(x_i) = 0,$$

i.e., x_i is a double root of P for all $i = 0, 1, \dots, n$. Hence P has $2(n+1) = 2n+2$ number of roots, and hence the Fundamental theorem of algebra yields that P is identically equal to 0, since the degree of P is at most $(2n+1)$. This implies that if the solution of the Hermite interpolation problem exists, it has to be unique.

Proof cont.

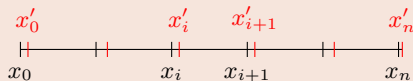
Direct computation gives that

$$H_{2n+1}(x_0) = f(x_0) \quad \text{and} \quad H'_{2n+1}(x_0) = f[x_0, x_0] = f'(x_0).$$

Next we show that

$$H_{2n+1}(x_1) = f(x_1) \quad \text{and} \quad H'_{2n+1}(x_1) = f'(x_1)$$

hold too. To prove this select numbers $x'_i > x_i$ close to x_i



so that the numbers

$$\{x_i, x'_i : i = 0, 1, \dots, n\}$$

be pairwise different, and let L_{2n+1} be the Lagrange polynomial interpolating the function values of f at these mesh points.

Proof cont.

Then

$$\begin{aligned}
 L_{2n+1}(x) = & f[x_0] + f[x_0, x'_0](x - x_0) + f[x_0, x'_0, x_1](x - x_0)(x - x'_0) \\
 & + f[x_0, x'_0, x_1, x'_1](x - x_0)(x - x'_0)(x - x_1) + \cdots \\
 & + f[x_0, x'_0, x_1, x'_1, \dots, x_n, x'_n](x - x_0)(x - x'_0) \cdots (x - x_{n-1}) \\
 & \cdot (x - x'_{n-1})(x - x_n),
 \end{aligned}$$

and

$$f(x) = L_{2n+1}(x) + f[x_0, x'_0, \dots, x_n, x'_n, x](x - x_0)(x - x'_0) \cdots (x - x_n)(x - x'_n).$$

The definition of L_{2n+1} and H_{2n+1} and the continuity of the divided difference yield for all x that

$$L_{2n+1}(x) \rightarrow H_{2n+1}(x) \quad \text{as } (x'_0, x'_1, \dots, x'_n) \rightarrow (x_0, x_1, \dots, x_n), \quad (5)$$

and so

$$f(x) = H_{2n+1}(x) + f[x_0, x_0, x_1, x_1, \dots, x_n, x_n, x](x - x_0)^2(x - x_1)^2 \cdots (x - x_n)^2.$$

This proves relation (4).

Proof cont.

It follows from the uniqueness of the Lagrange polynomial that if we interchange x_0, x'_0 and x_1, x'_1 , then the interpolating polynomial remains the same, so

$$\begin{aligned}
 L_{2n+1}(x) &= f[x_1] + f[x_1, x'_1](x - x_1) + f[x_1, x'_1, x_0](x - x_1)(x - x'_1) \\
 &\quad + f[x_1, x'_1, x_0, x'_0](x - x_1)(x - x'_1)(x - x_0) + \cdots \\
 &\quad + f[x_1, x'_1, x_0, x'_0, x_2, x'_2, \dots, x_n, x'_n](x - x_1)(x - x'_1)(x - x_0)(x - x'_0) \\
 &\quad \cdot (x - x_2)(x - x'_2) \cdots (x - x_{n-1})(x - x'_{n-1})(x - x_n).
 \end{aligned}$$

Proof cont.

But then taking the limit $(x'_0, x'_1, \dots, x'_n) \rightarrow (x_0, x_1, \dots, x_n)$ of both sides, and using relation (5), we get

$$\begin{aligned}
 H_{2n+1}(x) &= f[x_1] + f[x_1, x_1](x - x_1) + f[x_1, x_1, x_0](x - x_1)^2 \\
 &+ f[x_1, x_1, x_0, x_0](x - x_1)^2(x - x_0) \\
 &+ f[x_1, x_1, x_0, x_0, x_2](x - x_1)^2(x - x_0)^2 \\
 &+ f[x_1, x_1, x_0, x_0, x_2, x_2](x - x_1)^2(x - x_0)^2(x - x_2) + \dots \\
 &+ f[x_1, x_1, x_0, x_0, x_2, x_2, \dots, x_n, x_n](x - x_1)^2(x - x_0)^2(x - x_2)^2 \\
 &\quad \dots (x - x_{n-1})^2(x - x_n).
 \end{aligned}$$

But from this form it is clear that

$$H_{2n+1}(x_1) = f(x_1) \quad \text{and} \quad H'_{2n+1}(x_1) = f'(x_1).$$

In a similar manner we can show that

$$H_{2n+1}(x_i) = f(x_i) \quad \text{and} \quad H'_{2n+1}(x_i) = f'(x_i), \quad i = 2, 3, \dots, n.$$

Theorem:

Let $f \in C^{2n+2}$. Then there exists $\xi \in \langle x_0, x_1, \dots, x_n, x \rangle$ such that

$$f(x) - H_{2n+1}(x) = \frac{f^{(2n+2)}(\xi)}{(2n+2)!} (x - x_0)^2 \dots (x - x_n)^2.$$

Proof

Let x be a number different from all mesh points, and define the function

$$g(z) := f(z) - H_{2n+1}(z) - \frac{(z - x_0)^2 \dots (z - x_n)^2}{(x - x_0)^2 \dots (x - x_n)^2} (f(x) - H_{2n+1}(x)).$$

Clearly, $g \in C^{2n+2}$, and $x_0, \dots, x_n$ are all double roots, and x is a simple root of g . Therefore the generalized Rolle's Theorem implies that there exists $\xi \in \langle x_0, x_1, \dots, x_n, x \rangle$, such that

$$g^{(2n+2)}(\xi) = 0.$$

This yields the statement of the theorem.

Corollary

Suppose $f \in C^{2n+2}$ and $x, x_0, \dots, x_n$ are pairwise different numbers. Then there exists $\xi \in \langle x_0, x_1, \dots, x_n, x \rangle$ such that

$$f[x_0, x_0, \dots, x_n, x_n, x] = \frac{f^{(2n+2)}(\xi)}{(2n+2)!}.$$

Table of divided differences for the Hermite polynomial

x_0	$f(x_0)$				
x_0	$f(x_0)$	$f[x_0, x_0]$			
x_1	$f(x_1)$	$f[x_0, x_1]$	$f[x_0, x_0, x_1]$		
x_1	$f(x_1)$	$f[x_1, x_1]$	$f[x_0, x_1, x_1]$	$\ddots$	
$\vdots$	$\vdots$	$\vdots$	$\vdots$		
x_n	$f(x_n)$	$f[x_{n-1}, x_n]$	$f[x_{n-1}, x_{n-1}, x_n]$	$\cdots$	
x_n	$f(x_n)$	$f[x_n, x_n]$	$f[x_{n-1}, x_n, x_n]$	$\cdots$	$f[x_0, x_0, x_1, x_1, \dots, x_n, x_n]$

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

-1	-1					
-1	-1	-5				
1	1	×	×			
1	1	7	×	×		
2	29	×	×	×	×	
2	29	61	×	×	×	×

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

-1	-1					
-1	-1	-5				
1	1	1	×			
1	1	7	×	×		
2	29	×	×	×	×	
2	29	61	×	×	×	×

$$\frac{1 - (-1)}{1 - (-1)} = 1$$

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

-1	-1						
-1	-1	-5					
1	1	1	×				
1	1	7	×	×			
2	29	28	×	×	×		
2	29	61	×	×	×	×	

$$\frac{29-1}{2-1} = 28$$

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

-1	-1						
-1	-1	-5					
1	1	1	3				
1	1	7	×	×			
2	29	28	×	×	×		
2	29	61	×	×	×	×	

$$\frac{1 - (-5)}{1 - (-1)} = 3$$

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

-1	-1						
-1	-1	-5					
	1	1	1	3			
1	1	7	3	×			
2	29	28	×	×	×		
2	29	61	×	×	×	×	

$$\frac{7-1}{1-(-1)} = 3$$

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

-1	-1						
-1	-1	-5					
1	1	1	3				
1	1	7	3	×			
2	29	28	21	×	×		
2	29	61	×	×	×	×	

$$\frac{28-7}{2-1} = 21$$

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

-1	-1					
-1	-1	-5				
1	1	1	3			
1	1	7	3	×		
2	29	28	21	×	×	
2	29	61	33	×	×	×

$$\frac{61-28}{2-1} = 33$$

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

-1	-1					
-1	-1	-5				
1	1	1	3			
1	1	7	3	0		
2	29	28	21	×	×	
2	29	61	33	×	×	×

$$\frac{3-3}{1-(-1)} = 0$$

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

-1	-1						
-1	-1	-5					
1	1	1	3				
1	1	7	3	0			
2	29	28	21	6	2		
2	29	61	33	12	×	×	

$$\frac{6-0}{2-(-1)} = 2$$

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

-1	-1						
-1	-1	-5					
1	1	1	3				
1	1	7	3	0			
2	29	28	21	6	2		
2	29	61	33	12	2	0	

$$\frac{2-2}{2-(-1)} = 0$$

Example

Find the Hermite interpolating polynomial corresponding to the data:

x_i	-1	1	2
y_i	-1	1	29
y'_i	-5	7	61

We fill out the following table of divided differences:

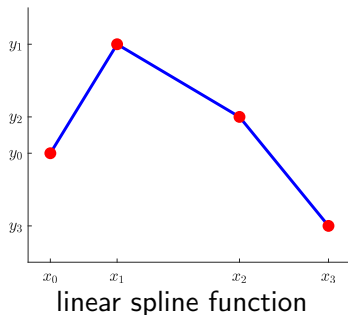
-1	-1					
-1	-1	-5				
1	1	1	3			
1	1	7	3	0		
2	29	28	21	6	2	
2	29	61	33	12	2	0

Therefore the Hermite polynomial is of degree 4, and

$$H_5(x) = -1 - 5(x+1) + 3(x+1)^2 + 2(x+1)^2(x-1)^2 = 2x^4 - x^2 + x - 1.$$

6.5 Spline Interpolation

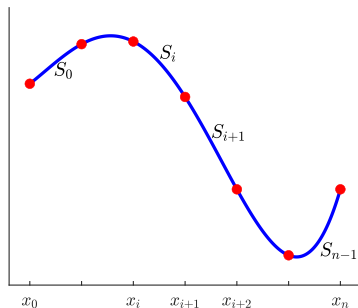
Let $a = x_0 < x_1 < \dots < x_n = b$ be a division of the interval $[a, b]$. The continuous function $S: [a, b] \rightarrow \mathbb{R}$ is a **spline function of degree k** corresponding to the mesh $\{x_i\}$ if $S \in C^{k-1}[a, b]$, and the restriction of S to each interval $[x_i, x_{i+1}]$ is a polynomial of degree at most k . The first, second and third-order spline functions are called **linear**, **quadratic** and **cubic spline functions**, respectively.



We investigate cubic spline interpolation.

Suppose given pairwise different mesh points $a = x_0 < x_1 < \dots < x_n = b$ and corresponding function values $y_0, y_1, \dots, y_n$. We are looking for a cubic spline function S which interpolates the given data, i.e., it satisfies

$$S(x_i) = y_i, \quad i = 0, 1, \dots, n.$$



The restriction of S to the interval $[x_i, x_{i+1}]$ is denoted by S_i ($i = 0, 1, \dots, n - 1$).

The functions S_i satisfy:

$$S_i(x_i) = y_i, \quad i = 0, 1, \dots, n-1, \quad (6)$$

$$S_i(x_{i+1}) = y_{i+1}, \quad i = 0, 1, \dots, n-1, \quad (7)$$

$$S'_i(x_{i+1}) = S'_{i+1}(x_{i+1}), \quad i = 0, 1, \dots, n-2, \quad (8)$$

$$S''_i(x_{i+1}) = S''_{i+1}(x_{i+1}), \quad i = 0, 1, \dots, n-2. \quad (9)$$

Since the polynomials S_i are defined by 4 parameters, so S is determined by $4n$ parameters. The number of conditions in (6)–(9) is only $4n - 2$, therefore this problem has no unique solution yet. Hence we add two more conditions:

$$S''_0(x_0) = 0 \quad \text{and} \quad S''_{n-1}(x_n) = 0. \quad (10)$$

A cubic spline function defined by conditions (6)–(10) is called **natural spline** function.

Consider the functions S_i in the form:

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3,$$

where a_i , b_i , c_i and d_i ($i = 0, 1, \dots, n-1$) are parameters to be determined. Then

$$\begin{aligned} S_i'(x) &= b_i + 2c_i(x - x_i) + 3d_i(x - x_i)^2, \\ S_i''(x) &= 2c_i + 6d_i(x - x_i). \end{aligned}$$

These equations imply

$$a_i = S_i(x_i) = y_i, \quad b_i = S_i'(x_i) \quad \text{and} \quad c_i = S_i''(x_i)/2, \quad i = 0, 1, \dots, n-1. \quad (11)$$

With the help of relation (11) we define the constants a_n , b_n and c_n :

$$a_n := y_n, \quad b_n := S'(x_n) \quad \text{and} \quad c_n := S''(x_n)/2. \quad (12)$$

(The derivatives in (12) denote left sided derivatives.) Substituting $x = x_{i+1}$ into the formula of S_i , and using equation (7) and relation $a_i = y_i$, we get

$$y_i + b_i(x_{i+1} - x_i) + c_i(x_{i+1} - x_i)^2 + d_i(x_{i+1} - x_i)^3 = y_{i+1}.$$

Introduce the notations $\Delta x_i := x_{i+1} - x_i$ and $\Delta y_i := y_{i+1} - y_i$. Then

$$b_i \Delta x_i + c_i (\Delta x_i)^2 + d_i (\Delta x_i)^3 = \Delta y_i, \quad i = 0, 1, \dots, n-1. \quad (13)$$

Condition (8) and relation $b_{i+1} = S'_{i+1}(x_{i+1})$ yield

$$b_i + 2c_i \Delta x_i + 3d_i (\Delta x_i)^2 = b_{i+1} \quad (14)$$

for $i = 0, 1, \dots, n-2$. Using the definition of b_n we get that (14) holds for $i = n-1$ too. Similarly, (9) and the definition of c_n imply

$$2c_i + 6d_i \Delta x_i = 2c_{i+1}, \quad i = 0, 1, \dots, n-1,$$

hence

$$d_i = \frac{c_{i+1} - c_i}{3\Delta x_i}, \quad i = 0, 1, \dots, n-1. \quad (15)$$

Substituting it back to equations (13) and (14) we get

$$b_i \Delta x_i + c_i (\Delta x_i)^2 + \frac{c_{i+1} - c_i}{3} (\Delta x_i)^2 = \Delta y_i, \quad i = 0, 1, \dots, n-1,$$

$$b_i + 2c_i \Delta x_i + (c_{i+1} - c_i) \Delta x_i = b_{i+1}, \quad i = 0, 1, \dots, n-1.$$

From the first equation we express

$$b_i = \frac{\Delta y_i}{\Delta x_i} - \frac{2c_i + c_{i+1}}{3} \Delta x_i,$$

and substituting it into the second equation for $i = 0, 1, \dots, n-2$ we get

$$c_i \Delta x_i + 2c_{i+1} (\Delta x_i + \Delta x_{i+1}) + c_{i+2} \Delta x_{i+1} = 3 \frac{\Delta y_{i+1}}{\Delta x_{i+1}} - 3 \frac{\Delta y_i}{\Delta x_i}. \quad (16)$$

Note that (16) holds for any cubic spline interpolation.

Equation (16) determines a system of $n-1$ linear equations for c_i . We add equations $c_0 = 0$ and $c_n = 0$ following from condition $S''_0(x_0) = 0$ and $S''_{n-1}(x_n) = 0$.

So we get a $n + 1$ -dimensional linear system of the form $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{x} = (c_0, c_1, \dots, c_n)^T$,

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & \dots & 0 \\ \Delta x_0 & 2(\Delta x_0 + \Delta x_1) & \Delta x_1 & 0 & 0 & \dots & 0 \\ 0 & \Delta x_1 & 2(\Delta x_1 + \Delta x_2) & \Delta x_2 & 0 & \dots & 0 \\ & & \ddots & \ddots & \ddots & & \\ 0 & \dots & & \Delta x_{n-2} & 2(\Delta x_{n-2} + \Delta x_{n-1}) & \Delta x_{n-1} \\ 0 & \dots & & 0 & 0 & 1 \end{pmatrix}$$

is a tridiagonal matrix and

$$\mathbf{b} = \begin{pmatrix} 0 \\ 3\frac{\Delta y_1}{\Delta x_1} - 3\frac{\Delta y_0}{\Delta x_0} \\ \vdots \\ 3\frac{\Delta y_{n-1}}{\Delta x_{n-1}} - 3\frac{\Delta y_{n-2}}{\Delta x_{n-2}} \\ 0 \end{pmatrix}.$$

Since $\mathbf{A}$ is diagonally dominant, the system $\mathbf{Ax} = \mathbf{b}$ has a unique solution.

Then with the help of c_i , we can compute the coefficients d_i and b_i .

Theorem:

The problem of natural cubic spline interpolation has a unique solution.

Example

Find the natural cubic spline interpolation of the given data:

x_i	0.0	0.5	1.5	2.0	3.0	4.0
y_i	0.9	0.1	1.5	0.0	-0.8	-0.2

Using the notations introduced before the linear system of the coefficients c_i is

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0.5 & 3 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 3 & 1 & 0 \\ 0 & 0 & 0 & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 9.0 \\ -13.2 \\ 6.6 \\ 4.2 \\ 0 \end{pmatrix}.$$

Example cont.

The resulting natural spline function is:

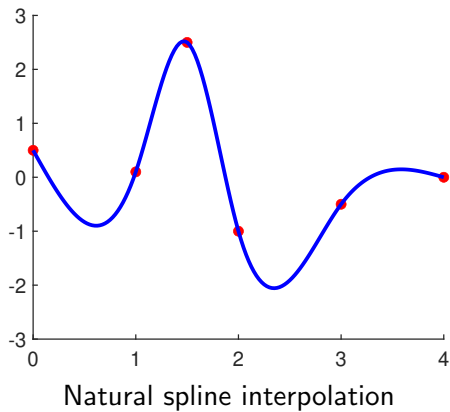
$$S_0(x) = 0.9 - 2.47117647x + 3.48470588x^3,$$

$$S_1(x) = 0.1 + 0.142352940(x - 0.5) + 5.22705882(x - 0.5)^2 \\ - 3.96941176(x - 0.5)^3,$$

$$S_2(x) = 1.5 - 1.31176471(x - 1.5) - 6.68117647(x - 1.5)^2 \\ + 6.60941177(x - 1.5)^3,$$

$$S_3(x) = -3.03588235(x - 2) + 3.232941176(x - 2)^2 \\ - 0.997058823(x - 2)^3,$$

$$S_4(x) = -0.8 + 0.438823529(x - 3) + 0.2417647059(x - 3)^2 \\ - 0.0805882353(x - 3)^3.$$



Instead of condition (10) we can specify other boundary conditions for S . Now we investigate condition

$$S'(x_0) = y'_0 \quad \text{and} \quad S'(x_n) = y'_n, \quad (17)$$

where y'_0 and y'_n are given numbers. This means that we know (specify) the slope of the tangent line of S at the end points of the interval. A cubic spline which satisfy conditions (17) is called **clamped spline** function.

Theorem:

Let $a = x_0 < x_1 < \dots < x_n = b$ be mesh points and $y_0, y_1, \dots, y_n$ be function values, and let S be the natural cubic spline interpolating function associated to the given data. Then

$$\int_a^b (S''(x))^2 dx \leq \int_a^b (f''(x))^2 dx$$

for every $f \in C^2[a, b]$, which also interpolates the given data, i.e.,

$$f(x_i) = y_i, \quad \text{for } i = 0, 1, \dots, n.$$

Proof

Introduce the function

$$g(x) := f(x) - S(x).$$

Then

$$f''(x) = S''(x) + g''(x),$$

and so

$$\int_a^b (f''(x))^2 dx = \int_a^b (S''(x))^2 dx + 2 \int_a^b S''(x)g''(x) dx + \int_a^b (g''(x))^2 dx.$$

Proof cont.

Since $\int_a^b (g''(x))^2 dx \geq 0$, the statement of the theorem follows if we show

$$\int_a^b S''(x)g''(x) dx = 0.$$

Using integration by parts we get

$$\begin{aligned} \int_a^b S''(x)g''(x) dx &= \sum_{i=1}^n \int_{x_{i-1}}^{x_i} S''(x)g''(x) dx \\ &= \sum_{i=1}^n [S''(x)g'(x)]_{x_{i-1}}^{x_i} - \sum_{i=1}^n \int_{x_{i-1}}^{x_i} S'''(x)g'(x) dx \\ &= S''(b)g'(b) - S''(a)g'(a) - \sum_{i=1}^n \int_{x_{i-1}}^{x_i} S'''(x)g'(x) dx. \end{aligned}$$

Since S is a natural spline function, we have $S''(a) = S''(b) = 0$. Since S is a third-order polynomial over the intervals $[x_{i-1}, x_i]$, its third derivative is constant, which can be factored out in front of the integral. But

$$\int_{x_{i-1}}^{x_i} g'(x) dx = g(x_i) - g(x_{i-1}) = 0, \text{ since } g(x_i) = 0 \text{ for } i = 0, 1, \dots, n.$$

Theorem:

Let $f \in C^4[a, b]$, $a = x_0 < x_1 < \dots < x_n = b$ mesh points, $y_i = f(x_i)$, $i = 0, 1, \dots, n$ function values, and $y'_0 = f'(a)$ and $y'_n = f'(b)$ derivative values, and let S be the corresponding clamped cubic spline function. Then for $x \in [a, b]$ it follows

$$|f(x) - S(x)| \leq \frac{5}{384} M_4 h^4,$$

$$|f'(x) - S'(x)| \leq \left(\frac{\sqrt{3}}{216} + \frac{1}{24} \right) M_4 h^3,$$

$$|f''(x) - S''(x)| \leq \left(\frac{1}{12} + \frac{h}{3k} \right) M_4 h^2,$$

where $M_4 := \max\{|f^{(4)}(x)| : x \in [a, b]\}$,
 $h := \max\{x_{i+1} - x_i : i = 0, 1, \dots, n-1\}$,
 $k := \min\{x_{i+1} - x_i : i = 0, 1, \dots, n-1\}$.